LESSON PLAN

LSF, STABILITY, A/C PERFORMANCE (6)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	32
Total No. of Lesson Plans	33	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Low speed wind tunnel and wind tunnel balances
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Low speed wind tunnel and wind tunnel balances. Demonstration on pressure plotting experiments
5. Specific Objectives	6.1.5 .1Trainee will be able to explain about wind tunnel operation

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration on pressure plotting experiments; Safety: Ensure model is securely clamped; Instructor supervision: Monitor balance readings and data recording
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a wind tunnel?
(b) Name two types of wind tunnels used in aerodynamics.
(c) Why is a low speed wind tunnel preferred for subsonic testing?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Components of Low Speed Wind Tunnel • Operation of Wind Tunnel Fan System • Types of Wind Tunnel Balances • Principle of Force Measurement • Calibration of Balance Systems	L & Ds	PPT, IB	15 min
(b)	• Demonstration of Pressure Plotting • Setup of Balance and Model • Recording of Lift and Drag Data • Analysis of Pressure Distribution • Interpretation of Force Coefficients	L & Ds	PPT, PDF	20 min
(c)	• Sources of Experimental Error • Impact of Blockage Ratio • Tunnel Wall Interference Effects • Data Validation Techniques • Reporting of Experimental Results	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how a strain gauge balance measures lift and drag.
(b) Describe the effect of blockage ratio on wind tunnel test results.
(c) How does pressure plotting help in determining aerodynamic forces?

12. Home Assignment and Follow Up	Prepare a brief note on the procedure to be followed for the next lesson regarding measurements of lift and drag in a wind tunnel, referencing LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Measurements of lift and drag in wind tunnel.	Refer to the master syllabus reference list	LP 33

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min